

Animal Linguistics as an Integrative Framework for Animal Communication*

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March 28, 2026

Abstract: Interest in animal communication has arguably reached an all-time high, with two very visible projects—the search for the animal roots of human language and the development of methods and technologies to decode animal signals. These projects need a shared, overarching scientific framework. We argue that animal linguistics offers one, which will require close collaboration between ethologists, linguists and, increasingly, AI specialists. With the same ambition as human linguistics, animal linguistics seeks to develop explicit models of animal communication ("formal animal linguistics"), to understand the extent and limits of variation in living species ("comparative animal linguistics"), and to investigate the cognitive basis of observed properties ("animal psycholinguistics"). Through a "phylogenetic tree of animal signals", it seeks to retrace and explain the path from ancestral signals to modern ones ("evolutionary animal linguistics"). In its synchronic and diachronic components, animal linguistics offers an integrative framework for scientific and technological progress in animal communication. Going forward, it will offer new areas of application and new sources of insight for multiple areas of cognitive science.

Keywords: animal linguistics, animal communication, bioacoustics, animal signals, evolution of communication, evolutionary animal linguistics

Word count with tables: 3,918 words

1 Why animal linguistics now

Interest in animal communication has arguably reached an all-time high, both among scientists and the general public. The most visible contemporary projects aim to discover the evolutionary roots of human language ([4][125][42][44][113][114][21][120]), or to develop methods and AI-based technologies to understand animals ([89][23][43][33][87][58]). But the field lacks a shared, overarching scientific framework. We argue that animal linguistics offers one, which will require close collaboration between ethologists, linguists and, increasingly, AI specialists (see also the

* **Acknowledgments:** We are very grateful to Jeffrey Heinz and Robin Ryder for helpful discussions and references.

Funding:

Schlenker: This research received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (grant agreement No 788077, Orisem, PI: Schlenker). Research was conducted at DEC, Ecole Normale Supérieure - PSL Research University, supported by grants FrontCog ANR-17-EURE-0017 and ANR-10-IDEX-0001-02. Mathevon: This research received funding from the Institut universitaire de France.

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sister program of animal language processing [52][40]). Animal linguistics builds on decades of work on bioacoustics and animal signals ([74][88]), and seeks to develop formal models of animal communication that are as explicit as those of human linguistics ([97][14][113][116]). Its program has two components. In its synchronic part ([97]), it investigates current nonhuman animal communication systems as formal systems (“formal animal linguistics”), organizes their properties within a systematic typology (“comparative animal linguistics”), and investigates their cognitive basis (“animal psycholinguistics”). In its diachronic part ([100]), it reconstructs with phylogenetic methods ancestral signals that were used millions of years ago, organizes them into an evolutionary tree of animal signals, and builds models to understand their laws of evolution (“evolutionary animal linguistics”).

The synchronic part of animal linguistics is crucial to develop benchmarks to assess successes in decoding animal communication with AI technologies and large corpora ([87][77][85]), just like theoretical linguistics has helped develop benchmarks for natural language processing technologies ([67][72][53][57]). The diachronic part of animal linguistics is essential to assess the strength of purported continuities with human language, and their likely status as deriving from convergent evolution or common descent: any continuities must be assessed in a broader typological and evolutionary background. Overall, a unifying framework is key to properly describing both similarities and differences among the various systems of communication in nature.

2 Synchronic animal linguistics offers an integrative framework for extant animal communication systems

In linguistics, the adoption of methods from logic and formal language theory in the 1950's-1960's ([31][78][79]) made it possible to (i) state theories with complete clarity, (ii) derive testable predictions from them, (iii) ask about their cognitive underpinnings, and (iv) offer a theoretically informed typology of human languages: formal linguistics is by nature comparative ([32][102][7]). The same motivations apply to animal linguistics in its formal, comparative and psycholinguistic components.

(1) The main components of animal linguistics

Component	Definition	Illustrations
Formal Animal Linguistics	It develops explicit and predictive models of the properties of a given communication system.	Definition of the signals used by a species, specification of their meanings, specification of rules of combination of signals and corresponding meanings – e.g. in Japanese tits, ABC as an alarm call, D as a recruitment call and ABC-D as a combination yielding a mobbing-related meaning.
Comparative Animal Linguistics	Using the results of Formal Animal Linguistics for diverse communication systems, it delivers a typology of their properties.	Comparison of signal functions across species— e.g. Japanese tits and chimpanzee have sequences of alarm and recruitment calls that have related functions.
Animal Psycholinguistics	It investigates (i) the cognitive abilities that	(i) Cognitive abilities: chimpanzees take into account the information state of their audience when producing

	underlie the properties uncovered by Formal and Comparative Animal Linguistics, and (ii) their acquisition path.	snake alarm calls (“do not state the obvious”). This may require some ability to represent other minds. (ii) Acquisition path: vervet monkeys initially overapply eagle alarm calls, but only to birds.
Evolutionary Animal Linguistics	It reconstructs ancestral communication systems, places them on the phylogenetic tree of species, and investigates their laws of evolution.	Phylogenetic tree of the waggle dance of bees, originating 20 million years ago and giving rise to horizontal and vertical dancing in different species. Investigation of the original dance, why it arose, and why it evolved.

To sidestep definitional issues (“are you allowed to call x a language?”), it is expedient to adopt terminology from formal language theory ([97]). Any set of strings can be called a language, and the interesting question is what its properties are. Similarly, to sidestep issues such as “what is meaning”, one can adopt the classic view, from linguistics ([55][94]), that the meaning of an expression is equated to the situations that make it true (or accurate, or appropriate, etc.).

Linguistics divides communicative labor among different modules—phonology, morphology, syntax, semantics, pragmatics. This division follows from conceptual considerations about communicative systems in general ([97][14][98]). Applying it to animal systems doesn't imply that the specific mechanisms of human linguistics are applicable to animals, but it offers an efficient way to assess similarities and differences.

Phonological units are the cognitive atoms (= smallest relevant units) of linguistic form. In human linguistics, they were traditionally considered to be elementary sound types with dedicated time slots (phonemes), somewhat reminiscent of letters. They are now thought to be features ([54]), corresponding for instance to articulatory instructions (e.g. whether the vocal folds vibrate or not); phonemes are thus feature bundles. Some features do not have a dedicated time slot, e.g. tones that make up question intonation ([66]). The feature-based view is not tied to the vocal modality and extends to manual and facial articulators in sign language phonology ([91]).

Phonological units are usually devoid of meaning. They can be combined in rule-governed ways to create meaningful elements, morphemes—e.g. *red*, *green*, *ish*. **Morphemes** can be combined to form words (by rules of morphology)—e.g. *green+ish*→*greenish*. And words are combined by rules of **syntax** to form sentences, e.g. *This is greenish*.

To model meaning, **semantics** associates truth conditions to sentences through lexical definitions and rules of meaning combination, and **pragmatics** investigates the interaction between meaning and context (e.g. the speaker's communicative goals). In all areas, **psycholinguistics** investigates the cognitive abilities that underlie the formal properties observed ([83][94][19]).

Building on bioacoustics ([6][74]), “**animal phonology**” investigates the cognitive atoms of signal form ([121][20][42][34][13]). Concretely, this starts with the development of a minimal yet lossless transcription. It may for instance involve concatenable calls, or notes, or features, some of which may span several time slots (e.g., certain melodic slopes encode species identity, individual identity, or distress [73][5][99]). Things are easier when these units are meaningful (and thus morphemes) because this establishes their “cognitive relevance”, but they do not have to be. For instance, chestnut-crowned babblers have AB flight calls that coordinate group movement, and BAB prompt/provisioning calls that stimulate nestling begging. A and B are apparently

meaningless on their own, hence possibly not morpheme-like but phoneme-like ([41][20]). Vowel-like elements have been found in sperm whales, but without proof that they can be combined into meaningful elements ([13]; see also [84]). In birdsongs, notes without apparent meaning have also been found to combine through sophisticated rules ([9][22]). Interestingly, birdsongs usually involve patterns found in formal languages described as “regular”, a property shared with human phonology ([16][56]). This motivates a “phonological continuity hypothesis” ([45]), according to which signal sequences in animals involve rules found in human phonology, not in human syntax.

At the morphological level (“**animal morphology**”), signals may be simple or (rarely) complex. In Campbell’s monkeys ([36]), the simplex calls *krak* and *hok* are used for general alerts and non-ground alerts respectively. In addition, a suffix *-oo* can yield the complex calls *krak-oo* and *hok-oo*, modifying the meaning of *krak* and *hok* in roughly the way *-ish* modifies *green* and *red* ([80][95][65][35]).

“**Animal syntax**” is concerned with combinations of meaningful signals ([124][125][62][63]). While signal sequences are very common, a key question is whether ordering restrictions result from specifically syntactic rules or from broader cognitive or physiological constraints. Campbell’s monkeys have non-predation *boom* calls which always appear at the beginning of sequences. This may be seen as a rule of syntax ([122]), or possibly as reflecting production constraints, as *booms* require that the monkey fill air sacs ([97]).

On the **meaning side** (“**animal semantics**”), one must specify the conditions under which the smallest meaningful units are true. When they are combined, whether by rules of morphology or syntax, one must specify how the meaning of the whole is obtained. A key question is whether there are animal instances of compositionality, whereby the meaning of the whole is derived from the meaning of the parts (as in *very hot*; [64]). At the morphological level, a compositional rule was posited for Campbell’s monkeys’ *-oo* suffix, mentioned above. At the syntactic level, there are undoubtedly trivial instances of compositionality when calls are combined as separate utterances. Thus in titi monkeys, a series of A-calls can be followed by a series of B-calls, but it simply yields the conjunctions of their meanings, just like when Ann says *It’s hot* and Bill says in roughly the same place *It’s humid*: one understands that it’s both hot and humid ([98]). The key question is whether there are non-trivial instances of compositionality in nonhumans (as in the case of *very hot*). One striking case was motivated for the ABC-D call of Japanese tits: ABC means “alert”, D means “help is needed”, and ABC-D serves to attract other birds to help mob a predator. Crucially, ABC-D behaves differently from separate utterances of ABC and D (e.g. as played by different loudspeakers), but its meaning is arguably derived from that of ABC and D ([115][114][98][101]). Remarkably, some researchers now argue that non-trivial compositionality in this sense is found in large corpora of bonobo and chimpanzee calls ([15][50]).

Unsurprisingly, meaning interacts with the environmental context. When titi monkeys hear an A-call (“serious non-ground threat”), they should think of raptors because they know that they are predators. A second source of enrichment involves competition among calls (“animal pragmatics”). When there is a raptor, titis could in principle produce A-calls (“serious non-ground threat”) or B-calls (“general alert”), but they always start by producing A-calls. This makes sense if they use the most informative calls available (“Informativity Principle” [97][96]). On this assumption, the B-call is only used for alerts *other* than serious non-ground ones, hence an enriched meaning. While the existence of such competition principles requires further investigation ([112]), it offers powerful new analytical tools. Beyond competition principles, meaning can be enriched by reconstructing the intentions of the signaler. This is a standard property of human language, central in multiple non-literal uses of language—humans

communicate by making their intentions overt ([51]), but this is thought to be beyond animals' abilities ([104]).

These levels are summarized in Table (2), with their key properties, usually sharply distinct from human counterparts, and salient hypotheses. In all areas, animal psycholinguistics investigates the cognitive abilities that underlie the properties uncovered by formal animal linguistics and their acquisition path. For instance, chimpanzees tend to produce snake alarm calls for interlocutors that have not seen the threat, following a principle ('Do not state the obvious') that might require some representation of other minds ([37][107][118]), and infant vervet monkeys overapply eagle alarm calls, but only to birds ([106][108]).

(2) Levels of organization of animal linguistics, compared with human linguistics

Domain of communication	Example	Properties and hypotheses in animals	Properties and hypotheses in humans
Phonology <i>investigates the smallest (cognitively relevant) units</i>	-Animal calls: AB vs. BAB calls in chestnut crown babblers [41] -'Phonology' without associated meanings: birdsongs [16]	Very limited Birdsong hypothesis: birdsongs are subregular languages. [16]	Subregular languages [56]
Morphology <i>investigates the smallest meaningful units</i>	-oo suffix in Campbell's monkeys [80]	Very limited: few complex calls to date	Regular language [59]
Syntax <i>investigates the rules of combination of meaningful units</i>	-Trivial combinations: sequences of A and then B calls in titi monkeys [96] -Non-trivial combinations: ABC-D sequences in Japanese tits [114]	Very limited: few cases of non-trivial syntactic combinations Fitch's 'phonological syntax' hypothesis: Animal signals can be combined using the same rule types as in human phonology, not human syntax. [45] Rizzi's 1-merge hypothesis: Animals can sometimes merge signals, but non-recursively. [86]	Beyond context-free [82] Recursive merge + movement [111]
Semantics <i>investigates the rules deriving the meaning of simple and complex elements</i>	-Elementary signals: titi monkey A, B [96]; Japanese tit ABC, D [114] -Elementary acoustic features: melodic slope, nonlinearities [6] -Non-trivial composition: ABC-D sequences in Japanese tits [114]	Very limited: few cases of non-trivial semantic composition Default hypothesis: Signals are combined as separate utterances. [92][98]	Very rich (subpart of type theory in logic) [47] [119]
Pragmatics <i>investigates the interaction between meaning and context (e.g.</i>	-Informativity Principle: Use the most informative signal available [97] -Economy Principle: Do not state the obvious [94]	-The Informativity Principle has been posited in several species but needs further investigation. [97] -Chimpanzees might follow the Economy Principle. [37]	-Informativity and further principles of competition [93] -Different versions of the Economy Principle [60]

<i>communicative goals)</i>		Scott-Phillips and Heintz's hypothesis: Unlike humans, animals do not communicate by making their intentions overt. [104]	-Generalized enrichment on the basis of intentions [51]
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While similarities between animal and human languages are controversial, similarities *across* animal languages are striking, hence a comparative component, which seeks to determine the extent and limits of variability in animal communication systems. Cross-species similarities may be explained by common descent, by convergent evolution, or by coevolution for species that interacted over evolutionary periods. All three reasons are evolutionary in nature, and thus comparative animal linguistics leads into evolutionary animal linguistics.

3 Diachronic animal linguistics offers an integrative framework for the evolution of communication systems

Evolutionary animal linguistics uses phylogenetic information and methods to place animal signals on the tree of species, and it investigates evolutionary models that can explain observed regularities.

3.1 *Evolutionary tree of animal signals*

When DNA-based methods weren't widely available to reconstruct phylogenies, vocalization similarity was often used to reconstruct them because vocalizations were thought to be typically well-preserved ([25][49][49]). Turning the problem on its head, we can use well-established phylogenies to reconstruct the evolutionary history of animal signals ([90][117]). Strikingly, ancestral signals can be reconstructed over millions of years (by contrast, the state of human language a few hundred thousand years ago is controversial [30][44]). As an example, ABC and D calls are found throughout the family of Japanese tits (the Paridae family), and even combinations of ABC-D (in that order) likely existed in their most recent common ancestor, which lived 11 million years ago around Tibet ([90]). In further striking results, multiple ape gestures have been argued to be part of an innate repertoire that is more than 16 million years old ([24]). Systematizing these results into a tree of animal signals is an exciting collective enterprise for the field (with no assumption whatsoever that we will find a single ur-signal, as communication likely arose independently multiple times).

3.2 *Models of signal evolution*

Ancestral signal reconstruction should help inform evolutionary models of animal signals ([110]), which build on diverse fields. Evolutionary signaling theory has been particularly concerned with the emergence of signals, and their ability to convey truthful (honest) information ([75][105]); it has often relied on game-theoretic methods. For its part, bioacoustics has investigated detailed ways in which signal form is optimized for signal function (e.g. [6][70]). For instance, species information conveyed by white-browed warbler songs (conveyed by a slow descending modulation frequency) is heard far, as is needed if males are to attract conspecific females and repel conspecific males. By contrast, individual identity information is optimized for short-range communication ([73]), intended for close neighbors.

How do signals emerge and evolve? A key idea, made precise with evolutionary game-theoretic methods ([100]), is that the very first signals should have trouble arising. Suppose a group

of residents has no signal, and a mutant starts using one. This has a small cost (e.g. the mutant might become more conspicuous to predators), but no benefit, as the residents don't understand it, so the mutant's fitness should be low. While there are multiple solution strategies, a popular one is to assume that signals are understood from the start—maybe imperfectly, but enough to have an effect ([75][103][100]). This may for instance explain why many alarm calls contain nonlinearities, which are intrinsically hard to habituate to, and aversive ([17][1]).

Beyond the very first signals, one should explain the full sequence of development of signals in evolutionary times. For instance, if there are specific and general signals (like the A- vs. B-calls in tits), a key question is which came first, and why.

The development of a tree of animal signals should provide crucial constraints for evolutionary models. In addition, evolutionary models could explain how the same forces can give rise to similar signal types and even combinations in unrelated species. For instance, chimpanzees have an alarm-recruitment sequence that is surprisingly similar to the ABC-D mobbing sequence in Japanese tits ([68]). The chimpanzee sequence might initially be a candidate for a compositional sequence found in the common ancestor of humans and chimpanzees. But the point might be moot if strong evolutionary forces made such sequences appear multiple times. This is just one case among many in which apparent continuities between animals and humans must be assessed in the background of comparative and evolutionary animal linguistics, with a broad spectrum of distant species to assess convergent evolution.

4 Animal linguistics and cognitive science

While strongly anchored in ethology, animal linguistics requires expertise from multiple fields of cognitive science. Correspondingly, it offers these fields new opportunities of growth. To illustrate, questions about animal meanings turn out to be both conceptually and formally complex, sometimes devilishly so. To argue that Japanese tits or chimpanzees have or do not have non-trivial compositionality, the competing theories must be stated in great formal detail, and clear behavioral predictions must be derived, with an interplay between semantic, pragmatic and reasoning considerations—typically the kind of expertise that semanticists can bring to the table ([97][64]). Similarly, the detailed reconstruction of ancestral animal communication systems benefits from modeling and statistical methods used in historical linguistics ([8][117]). On the psycholinguistic side, infant studies are of direct relevance to animal studies, for two reasons: the methodological constraints are overlapping (no explicit instructions!); and the topics are similar, pertaining to cognitive correlates that may or may not be reflected in signals, in production or comprehension.

It is now abundantly clear that progress on animal communication cannot come from experiments alone, as these are extraordinarily labor-intensive. They must be complemented with large corpus data, and they must thus rely on recent methods of natural language processing (e.g. for large samples of audio signals or for multimodal data)—hence the term of "animal language processing" ([52]; see also [40]). Further examples of expected contributions of the language sciences to animal linguistics are listed in (3). Beyond the language sciences, expertise from cognitive science (including from animal cognition) will be crucial to understand the interaction between animal communication and other modules of cognition (such as perception and reasoning), or issues of brain implementation (within neuroscience).

(3) Potential contributions of different fields of the language sciences to animal linguistics

Fields	Contributions	Illustrations
Philosophy of language and mind	It will explore the relation between language, mind and reality in much simpler systems than human language.	-Constraints on possible concepts could apply to animal signals as well (e.g. [48][29]). -The Gricean approach to meaning in human language is entirely based on mind-reading abilities. These may be absent in animal species, hence new questions for the foundations of communication (e.g. [10][11][12][2][104]).
Philosophical logic and formal philosophy	-They will explore the formal properties of far simpler logical systems than are found in human language. -The models of meaning emergence proposed in formal philosophy will benefit from rich animal data.	The emergence of meaning in a traditional question of game-theoretic approaches to convention ([69]). Animal data (including based on ancestral meaning reconstruction) provide new constraints on evolutionary game-theoretic models (e.g. [110]).
Phonetics and phonology	-Phonetics will explore the articulatory and perceptual bases of animal signals, notably in the vocal and visual modalities. -When there is evidence that signals are made of smaller meaningless parts, phonology will explore their cognitive representation.	-In primates, production limitations might coexist with a 'language-ready' vocal apparatus, which might call for neural approaches to the production bottleneck ([44][46][18]). -Some sperm whale calls are reminiscent of phonemes ([13]). -A key issue in the cognitive analysis of multiple animal signals is their gradience, which might call for new theoretical tools (e.g. [61][109]). -Clear cases of sophisticated combinations might involve meaningless signals and finite-state rules (e.g. [16]).
Morphosyntax	It will explore the properties of animal signal combinations and their limitations.	-Arguments for or against non-trivial combinations require a detailed analysis (e.g. [98]). -Limitations of attested combinatorial properties might call for new typologies (e.g. [86]).
Semantics and pragmatics	-They will explore the properties of animal signal meanings and their combinations, as well as their limitations. -They will explore the division of labor between signal meaning, contextual knowledge and competition among signals.	-Detailed case studies involve a complex interaction between the properties of signal meanings, contextual knowledge and signal competition (e.g. [97]). -Recent analyses of bonobo calls make use of methods from distributional semantics ([15][28]).

Psycholinguistics	-Studies of animal communication share key constraints with infant studies and will benefit from their methods—e.g. to find indirect ways to provide instructions that cannot be given in natural language. -Topics are also overlapping, as they pertain to cognitive correlates or prerequisites of signals even when these can't be observed.	-Related methods are used to test compositionality in human infants and non-human primates ([38][39]). -In syntax, human infants and non-human primates display a preference for agents over patients, consistent with the prevalent word order pattern in languages ([21][71][76][81]).
Historical linguistics	It will contribute to methods of reconstruction for ancestral signals.	-Studies of the origin and evolution of song dialects benefits from techniques from human dialect studies ([117]). -Models of primate call evolution benefit from techniques from historical linguistics ([117]).
Natural language processing	Since field experiments are extraordinarily labor-intensive, analyses at scale will involve recent methods developed in natural language processing, be it for the analysis of large samples of audio signals or the joint analysis of multimodal data.	A new field is emerging under the name of "Animal Language Processing" ([52]; see also [1]).

Animal linguistics will not just benefit from cognitive expertise; it will also bring new light to several areas of cognitive science. First, it will help address the perennial question of the animal origins of human language: any claims of continuities can only be assessed in the background of the typology and evolution of animal communication systems, based on extensive data and detailed analyses. Second, animal linguistics will contribute to the comparative program of cognitive science: independently of issues of evolution, our understanding of human language will gain if we can situate it precisely within communication systems in the animal world. Third, animal linguistics will provide a window into animal cognition, as signaling behavior is cognitively conditioned.

Animal linguistics will thus benefit from and contribute to entire areas of cognitive science. Multiple cognitive scientists could get involved, but it goes without saying that they should do so in collaboration with specialists of animal behavior, probably by way of targeted projects of mutual interest.

Rooted in a conceptually motivated typology of modules of communication (phonology, morphology, syntax, semantics, pragmatics), animal linguistics promises to offer an integrative framework for animal communication systems, a reconstruction of their ancestral forms, and principles that explain their evolution. It is the necessary foundation for the quest of the animal roots of human language, and it will provide crucial analytical tools to assess any future successes in using technology to understand animals. More broadly, it will offer new areas of application and new sources of insight for multiple areas of cognitive science, and it will serve as a bridge to ethology.

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